

GeoEcoAI: Multi-Temporal Remote Sensing Assessment Report

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1. Executive Summary & Dataset Truth

This report presents the deep learning semantic land-cover segmentation and Post-Classification Comparison (PCC) change detection results evaluated on the **SECOND (SEmantic Change detectiON Dataset - Yang et al., IEEE TGRS 2021)**. The analysis utilizes a ResNet-50 backbone with U-Net multi-scale skip connections across 7 distinct land-cover classes.

2. Temporal Land Cover Distribution & Net Dynamics

Land Cover Class	T1 Area (ha)	T2 Area (ha)	Net Change (ha)	Trend
Water Bodies	0.18	0.19	+0.01	EXPANSION
Ground / Bare Soil	0.12	0.11	-0.01	REDUCTION
Low Vegetation / Agriculture	0.01	0.01	-0.00	REDUCTION
Tree / Forest Canopy	0.00	0.00	+0.00	EXPANSION
Building / Urban Infrastructure	0.07	0.07	+0.00	EXPANSION
Playground / Sports Ground	0.01	0.01	+0.00	EXPANSION

3. Rule-Based Environmental Observations

- [INFO] Urban & Infrastructure Expansion:** An increase in the area classified as urban/built-up land was detected. This spatial trend indicates active infrastructure development or residential expansion. *(+0.00 ha (+0.40%) expansion)*
- [INFO] Hydrological Surface Dynamics:** An expansion in classified water body surface was detected. This may correspond to precipitation accumulation, reservoir filling, or seasonal inundation. *(+0.01 ha (+4.24%) expansion)*
- [WARNING] Agricultural & Low Vegetation Dynamics:** A reduction in classified agricultural/low vegetation area was detected. This observation indicates potential conversion to bare soil, harvesting cycles, or land conversion. *(0.00 ha (15.85%) reduction)*

4. Decision Support & Policy Recommendations

- [LOW] Farmland Preservation & Soil Health Assessment:** Agricultural Land & Soil Preservation - Trigger: Cropland/low vegetation loss of 15.85% exceeds alert threshold (8.0%)
- Monitor conversion of prime agricultural soil into non-agricultural use.
 - Promote sustainable cover cropping to prevent topsoil erosion on fallow ground.

5. Scientific Limitations & Conclusion

Limitations: This framework evaluates high-resolution aerial imagery (0.5-3m RGB). While architecturally extensible to Landsat-8/9 through the multispectral input adapter, current experimental metrics reflect optical aerial sensor characteristics. Causal attribution requires auxiliary socio-economic and meteorological datasets.